

NOISE, VIBRATION AND HARSHNESS [G1058328]

DIAGNOSIS AND TESTING

PRINCIPLE OF OPERATION

For a detailed description of Noise, Vibration and Harshness issues, refer to the Description and Operation section of the workshop manual. REFER to: [Noise, Vibration and Harshness](#) (100-04 Noise, Vibration and Harshness, Description and Operation).

INSPECTION AND VERIFICATION

1. Verify the customer's concerns by operating the vehicle to duplicate the condition.
 2. Visually inspect the vehicle to determine any obvious cause(s) of the concern(s).
 3. If the inspection reveals obvious causes that can be readily identified, repair as necessary.
 4. If the concern(s) remains after the inspection, determine the symptom(s) and refer to the Symptom Chart.
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HOW TO USE THIS DIAGNOSTIC PROCEDURE SECTION

- Noise, vibration and harshness (NVH) concerns have become more important as vehicles have become more sensitive to these vibrations. This section is designed as an aid to identifying these situations
 - The section provides diagnostic procedures based on symptoms. If the condition occurs at high speed, for instance, the most likely place to start is under High Speed Shake
 - The road test procedure will tell how to sort the conditions into categories and how to tell a vibration from a shake
 - A series of Road Test Quick Checks is provided to make sure that a cause is either pinpointed or eliminated
 - Name the condition, proceed to the appropriate section and locate the correct diagnosis. When the condition is identified, the job is partly done
 - Follow the diagnostic procedure as outlined
 - Quick Checks are described within the step, while more involved tests and adjustments are outlined in General Procedures
 - Always follow each step exactly and make notes to recall important findings later
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CUSTOMER INTERVIEW

The road test and customer interview (if available) provide information that will help identify the concerns and will provide direction to the correct starting point for diagnosis.

NVH problems usually occur in a number of areas:

- tires
- engine accessories
- suspension
- driveline
- air leakage (wind noise)
- squeaks and rattles
- heating ventilation and air conditioning (HVAC)
- electrical (e.g. motor noise)
- transmission
- engine

It is important, therefore, that an NVH concern be isolated into its specific area(s) as soon as possible. The easiest and quickest way to do this is to carry out the Road Test as outlined.

NOISE DIAGNOSTIC PROCEDURE

NON-AXLE NOISE

The five most important sources of non-axle noise are exhaust, tires, roof racks, trim and mouldings, and transmission.

Therefore, make sure that none of the following conditions are the cause of the noise before proceeding with a driveline tear down and diagnosis.

- Under certain conditions, the pitch of the exhaust may sound very much like gear noise. At other times, it can be mistaken for a wheel bearing rumble
- Tires, especially snow tires, can have a high pitched tread whine or roar, similar to gear noise. Radial tires, to some degree, have this characteristic. Also, any non-standard tire with an unusual tread construction may emit a roar or whine type noise
- Trim and mouldings can also cause whistling or a whining noise
- Clunk may be a metallic noise heard when the automatic transmission is engaged in reverse or drive, or it may occur when the throttle is applied or released. It is caused by backlash somewhere in the driveline
- Bearing rumble sounds like marbles being tumbled. This condition is usually caused by a damaged wheel bearing

NOISE CONDITIONS

- Gear noise is typically a howling or whining due to gear damage or incorrect bearing preload. It can occur at various speeds and driving conditions, or it can be continuous
- Chuckle is a particular rattling noise that sounds like a stick against the spokes of a spinning bicycle wheel. It occurs while decelerating from approximately 64 km/h (40 miles/h) and can usually be heard all the way to a stop. The frequency varies with vehicle speed

- Knock is very similar to chuckle, though it may be louder and occurs on acceleration or deceleration. The tear down will disclose what has to be corrected

Check and rule out tires, exhaust and trim items before disassembling the transmission to diagnose and correct gear noise.

The noises described under Road Test usually have specific causes that can be diagnosed by observation as the unit is disassembled. The initial clues are the type of noise heard on the road test and the driving conditions.

VIBRATION CONDITIONS

NOTE:

New Constant Velocity (CV) joints should not be installed unless disassembly and inspection revealed unusual wear.

Clicking, popping or grinding noises may be caused by the following:

- Cut or damaged CV joint boots resulting in inadequate or contaminated lubricant in the outboard or inboard CV joint bearing housings
- Loose CV joint boot clamps
- Another component contacting the rear drive half shaft
- Worn, damaged or incorrectly installed wheel bearing, suspension or brake component

Vibration at highway speeds may be caused by the following:

- Out-of-balance front or rear wheels
- Out-of-round tires
- Driveline imbalance
- Driveline run-out (alignment)

NOTE:

Rear drive half shafts are not balanced and are not likely to contribute to rotational vibration disturbance.

Shudder or vibration during acceleration (including from rest) may be caused by the following:

- Driveline alignment
- Excessively worn or damaged outboard or inboard CV joint bearing housing
- Excessively high CV joint operating angles caused by incorrect ride height. Check ride height, verify correct spring rate and check items under Inoperative Conditions
- Excessively worn driveshaft components

LEAKAGE CONDITIONS

1. Inspect the CV joint boots for evidence of cracks, tears or splits.
2. Inspect the underbody for any indication of grease splatter in the vicinity of the rear drive half shaft, outboard and inboard CV joint boot locations, which is an indication of CV joint boot or CV joint boot clamp damage.

3. Inspect the inboard CV joint bearing housing seal for leakage.

INOPERATIVE CONDITIONS

If a CV joint or rear drive half shaft pull-out occurs, check the following:

- suspension components for correct location, damage or wear
- bushings for wear
- subframe for damage
- bent or worn components
 - Stabilizer bar link
 - Left-hand rear suspension lower arm and bushing
 - Right-hand rear suspension lower arm and bushing
 - Rear wheel hub and rear drive half shaft

ROAD TEST

A gear-driven unit will produce a certain amount of noise. Some noise is acceptable and may be audible at certain speeds or under various driving conditions as on a newly paved blacktop road. The slight noise is in no way detrimental and must be considered normal.

The road test and customer interview (if available) provide information needed to identify the condition and give direction to the correct starting point for diagnosis.

1. Make notes throughout the diagnosis routine. Make sure to write down even the smallest piece of information, because it may turn out to be the most important.
2. Do not touch anything until a road test and a thorough visual inspection of the vehicle have been carried out. Leave the tire pressures and vehicle load just where they were when the condition was first observed. Adjusting tire pressures, vehicle load or making other adjustments may reduce the conditions intensity to a point where it cannot be identified clearly. It may also inject something new into the system, preventing correct diagnosis.
3. Make a visual inspection as part of the preliminary diagnosis routine, writing down anything that does not look right. Note tire pressures, but do not adjust them yet. Note leaking fluids, loose nuts and bolts, or bright spots where components may be rubbing against each other. Check the luggage compartment for unusual loads.
4. Road test the vehicle and define the condition by reproducing it several times during the road test.
5. Carry out the Road Test Quick Checks as soon as the condition is reproduced. This will identify the correct diagnostic procedure. Carry out the Road Test Quick Checks more than once to verify they are providing a valid result. Remember, the Road Test Quick Checks may not tell where the concern is, but they will tell where it is not.

ROAD TEST QUICK CHECKS

1. 24-80 km/h (15-50 miles/h): With light acceleration, a moaning noise is heard and possibly a vibration is felt in the front floor pan. It is usually worse at a particular engine speed and at a particular throttle setting during acceleration at that speed. It

may also produce a moaning sound, depending on what component is causing it. Refer to Tip-In Moan in the Symptom Chart.

2. Acceleration/deceleration: With slow acceleration and deceleration, a shake is sometimes noticed in the steering wheel/column, seats, front floor pan, front door trim panel or front end sheet metal. It is a low frequency vibration (around 9-15 cycles per second). It may or may not be increased by applying brakes lightly. Refer to Idle Boom/Shake/Vibration in the Symptom Chart.
3. High speed: A vibration is felt in the front floor pan or seats with no visible shake, but with an accompanying sound or rumble, buzz, hum, drone or booming noise. Coast with the clutch pedal depressed or shift control selector lever in neutral and engine idling. If vibration is still evident, it may be related to wheels, tires, front brake discs, wheel hubs or front wheel bearings. Refer to High Speed Shake in the Symptom Chart.
4. Engine rpm sensitive: A vibration is felt whenever the engine reaches a particular rpm. It will disappear in neutral coasts. The vibration can be duplicated by operating the engine at the problem rpm while the vehicle is stationary. It can be caused by any component, from the accessory drive belt to the torque converter which turns at engine speed when the vehicle is stopped. Refer to High Speed Shake in the Symptom Chart.
5. Noise/vibration while turning: Clicking, popping, or grinding noises may be due to a worn, damaged, or incorrectly installed front wheel bearing, rear drive half shaft or CV joint.
6. Noise/vibration that is road speed relative: This noise/vibration can be diagnosed independent of engine speed or gear selected (engine speed varies but torque and road speed remain constant). The cause may be a rear drive axle/differential whine.

ROAD CONDITIONS

An experienced technician will always establish a route that will be used for all NVH diagnosis road tests. The road selected should be reasonably smooth, level and free of undulations (unless a particular condition needs to be identified). A smooth asphalt road that allows driving over a range of speeds is best. Gravel or bumpy roads are unsuitable because of the additional road noise produced. Once the route is established and consistently used, the road noise variable is eliminated from the test results.

NOTE:

Some concerns may be apparent only on smooth asphalt roads.

If a customer complains of a noise or vibration on a particular road and only on a particular road, the source of the concern may be the road surface. If possible, try to test the vehicle on the same type of road.

VEHICLE PREPARATION

Carry out a thorough visual inspection of the vehicle before carrying out the road test. Note anything which is unusual. Do not repair or adjust any condition until the road test is carried out, unless the vehicle is inoperative or the condition could pose a hazard to the technician.

After verifying the condition has been corrected, make sure all components removed have been installed.

After a road test, it is sometimes useful to do a similar test on a lift.

When carrying out the high-speed shake diagnosis or engine accessory vibration diagnosis on a lift, observe the following precautions:

WARNING:

If only one drive wheel is allowed to rotate, speed must be limited to 55 km/h (35 miles/h) indicated on the speedometer since actual wheel speed will be twice that indicated on the speedometer. Speed exceeding 55 km/h (35 miles/h) or allowing the drive wheel to hang unsupported could result in tire disintegration, differential failure, constant velocity joint and drive half shaft failure, which could cause serious personal injury and extensive vehicle damage. Failure to follow these instructions may result in personal injury.

CAUTION:

The suspension should not be allowed to hang free. When the CV joint is run at a very high angle, extra vibration as well as damage to the seals and joints can occur.

The rear suspension lower arm should be supported as far outboard as possible. To bring the vehicle to its correct ride height, the full weight of the vehicle should be supported in the rear by floor jacks. REFER to: (100-02 Jacking and Lifting)

[Jacking](#) (Description and Operation),

[Lifting](#) (Description and Operation).

1. Raise and support the vehicle. REFER to: (100-02 Jacking and Lifting)
[Jacking](#) (Description and Operation),
[Lifting](#) (Description and Operation).
2. Explore the speed range of interest using the Road Test Quick Checks as previously described.
3. Carry out a coast down in neutral. If the vehicle is free of vibration when operating at a steady indicated speed and behaves very differently in drive and coast, a transmission concern is likely.

Note, however, that a test on the lift may produce different vibrations and noises than a road test because of the effect of the lift. It is not unusual to find vibrations on the lift that were not found in the road test. If the condition found on the road can be duplicated on the lift, carrying out experiments on the lift may save a great deal of time.

EXHAUST NEUTRALIZATION PROCEDURE

1. Raise vehicle on lift and slacken all exhaust fixings.
2. With all fixings loose, neutralize the exhaust system.
3. Tighten all fixings to correct torque, starting at the rear-most point working towards the front of the vehicle.

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSE	ACTION
High-speed shake	▪ Wheel end vibration ▪ Engine/transmission ▪ Driveline	GO to Pinpoint Test A.
Tip-in moan	▪ Air cleaner ▪ Power steering ▪ Powertrain ▪ Engine mounts ▪ Exhaust system	GO to Pinpoint Test B.
Idle boom/shake/vibration, or shudder	▪ Cable(s)/hoses(s) ▪ Intake air distribution and filtering system ▪ Engine mounts ▪ Exhaust system ▪ Belt/pulleys	GO to Pinpoint Test C.
Wheel end vibration analysis	▪ Suspension/rear drive halfshaft and CV joints ▪ Tires/wheels ▪ Wheel bearings ▪ CV joint boots	GO to Pinpoint Test D.
Non-axle noise	▪ Trim/mouldings ▪ A/C system ▪ Accessories	GO to Pinpoint Test E.

PINPOINT TESTS

NOTE:

These Pinpoint Tests are designed to take the technician through a step-by-step diagnosis procedure to determine the cause of a condition. It may not always be necessary to follow the chart to its conclusion. Carry out only the Pinpoint Test steps necessary to correct the condition. Then check operation of the system to make sure the condition is corrected.

After verifying that the condition has been corrected, make sure all components removed have been installed.

PINPOINT TEST A : HIGH-SPEED SHAKE

A1: NEUTRAL COAST

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
	1 Carry out the neutral coast test.
	Does the vibration disappear during the neutral coast test? Yes Check and install/re-balance wheels and tires and driveshaft. Check and install new engine/transmission mounts as necessary. Repeat Road Test as outlined. No GO to D1.

PINPOINT TEST B : TIP-IN MOAN

B1: CHECK THE AIR CLEANER

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
	1 Check the air cleaner. Check the air cleaner, inlet tube, outlet tube, resonators and all other components associated with the air induction system for correct installation and tightness of all connections.
	Are the components OK? Yes GO to B2. No Correct the condition. Repeat the Road Test as outlined.

B2: CHECK THE EXHAUST SYSTEM

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
	1 Carry out the exhaust system neutralizing procedure in this section.
	Is the exhaust system OK? Yes GO to B3. No Repair as necessary. Restore vehicle. Repeat the Road Test as outlined.

B3: CHECK THE POWER STEERING

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
	1 Remove the auxiliary drive belt and test for tip-in moan.
	Is the tip-in moan OK? Yes Repair the power steering as necessary. For additional information, refer to Section 211-00. No Check and install new engine/transmission mounts as necessary. Repeat Road Test as outlined.

PINPOINT TEST C : IDLE BOOM/SHAKE/VIBRATION/SHUDDER

C1: CHECK CABLE/HOSES

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
	1 Check the engine compartment for any component that may be grounding between the engine and body or chassis. Example: air conditioning (A/C) hoses.

PINPOINT TEST C : IDLE BOOM/SHAKE/VIBRATION/SHUDDER

C1: CHECK CABLE/HOSES

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
	Are the components OK? Yes GO to C2. No Correct the condition. Repeat the Road Test as outlined.

C2: CHECK THE COOLING RADIATOR

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
	1 Check the engine cooling radiator mountings and bushings for security and condition. Check the radiator installation for any component that may have a touch condition.
	Are the installation and bushings OK? Yes GO to C3. No Correct the condition. Repeat the Road Test as outlined.

C3: CHECK THE EXHAUST SYSTEM

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
	1 Carry out the exhaust system neutralizing procedure in this section.
	Is the exhaust system OK? Yes Check and install new engine/transmission mounts as necessary. Repeat Road Test as outlined. No Repair as necessary. Repeat Road Test.

PINPOINT TEST D : WHEEL END VIBRATION ANALYSIS

D1: INSPECT THE TIRES

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
	1 Inspect the tires. ▪ Raise and support the vehicle. REFER to: (100-02 Jacking and Lifting) Jacking (Description and Operation), Lifting (Description and Operation). ▪ Inspect the tires for: ▪ Correct tire size ▪ Tire/wheel compatibility ▪ Wear or damage ▪ Tire beads correctly seated
	Are the tires OK? Yes GO to D2. No Inspect the wheels. For additional information, refer to Section 204-00.

D2: INSPECT WHEEL BEARINGS

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
	1 Inspect the wheel bearings. For additional information, refer to Section 204-00.
	Are the wheel bearings OK? Yes GO to D3. No Repair as necessary. Repeat the Road Test as outlined.

D3: INSPECT THE CONSTANT VELOCITY (CV) JOINT BOOTS

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
	1 Inspect the CV joint boots. - Spin the rear tire by hand - Inspect for evidence of cracks, tears, splits or splattered grease
	Are the CV joint boots OK? Yes GO to D4. No Repair as necessary. Repeat the Road Test as outlined.

D4: INSPECT WHEEL AND TIRE RUNOUT

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
	1 Inspect the wheel and tire runout. Carry out the Wheel and Tire Check procedure. REFER to: Lifting (100-02 Jacking and Lifting, Description and Operation).
	Is the wheel and tire runout OK? Yes Balance the wheels and tires. Refer to the wheel balance equipment manufacturers instructions. No Repair as necessary. REFER to: Lifting (100-02 Jacking and Lifting, Description and Operation). Repeat the Road Test as outlined.

PINPOINT TEST E : NON-AXLE NOISE

E1: INSPECT VEHICLE TRIM

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
	1 Check the grille and trim mouldings to see if they are the source of the noise.
	Are the vehicle trim components causing the noise? Yes Install new trim or repair as necessary. For additional information, refer to Section 501-08. No GO to E2.

E2: CHECK THE A/C SYSTEM FOR NOISE

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
	1 Check the A/C system components for noise by turning the A/C system on and off.
	Is the A/C system causing the noise? Yes Diagnose the A/C system. REFER to: Lifting (100-02 Jacking and Lifting, Description and Operation). No GO to E3.

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
	1 Inspect any accessories for being the source of the noise. Example: grounding body-to-frame, antennas, visors, bug deflectors and fog lights?
	Are the accessories the cause of the noise? Yes Adjust, repair or install new accessories or fasteners as required. No Verify the customer concern.